

Technical Publication

Diesel Engine
12 V 4000 P91
16 V 4000 P91

Application group
4B

Maintenance Schedule
M050776/00E



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Le manuel devra être observé en vue d'éviter des incidents ou des endommagements pendant le service. Aussi recommandons-nous à l'exploitant de le mettre à la disposition du personnel chargé de l'entretien et de la conduite.

Modifications réservées.

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El manual debe tenerse presente para evitar fallos o daños durante el servicio, y, por dicho motivo, el usuario debe ponerlo a disposición del personal de mantenimiento y de servicio.

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Il manuale va consultato per evitare anomalie o guasti durante il servizio, per cui va messo a disposizione dall'utente al personale addetto alla manutenzione e alla condotta.

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Para evitar falhas ou danos durante a operação, os dizeres do manual devem ser respeitados. Quem explora o equipamento economicamente consequentemente deve colocá-lo à disposição do respectivo pessoal da conservação, e à disposição dos operadores.

Salvo alterações.

1 Preface

MTU maintenance concept

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance planning and ensures a high level of equipment availability.

The maintenance schedule is based on the load profile / load factor specified below. The time intervals at which the maintenance work is to be carried out and the relevant checks and tasks involved are average values based on operational experience and are therefore to be regarded as guidelines only. Special operating conditions and technical requirements may require additional maintenance work and/or modification of the maintenance intervals. In order to be authorized to carry out the individual maintenance jobs, maintenance personnel must have achieved a level of training and qualification appropriate to the complexity of the task in hand. The various Qualification Levels QL1 to QL4 reflect the levels of training offered in MTU courses and the contents of the tool kits required:

QL1: Operational monitoring and maintenance which can be carried out during a break in operation without disassembling the engine.

QL3: Maintenance work which requires partial disassembly of the engine.

QL4: Maintenance work which requires complete disassembly of the engine.

The maintenance schedule matrix normally finishes with extended component maintenance. Following this, maintenance work is to be continued at the intervals indicated.

The "Task" numbers stated in the list of jobs to be done/action to be taken indicate the relevant maintenance item. They provide a reference to the scope of parts required and appear on the labels of the appropriate replacement parts.

Notes on maintenance

Specifications for fluids and lubricants, guideline values for their maintenance and change intervals and lists of recommended fluids and lubricants are contained in the MTU Fluids and Lubricants Specifications A001061 and in the fluids and lubricants specifications produced by the component manufacturers. They are therefore not included in the maintenance schedule (exception: deviations from the Fluids and Lubricants Specifications). All fluids and lubricants used must meet MTU specifications and be approved by the relevant component manufacturer.

Amongst other items, the operator/customer must carry out the following additional maintenance work:

- Protect components made of rubber or synthetic material from oil. Never treat them with organic detergents. Wipe with a dry cloth only.
- Fuel prefilter:

The maintenance interval depends on how dirty the fuel is. The paper inserts in fuel prefilters must be changed every two years at the latest (Task 9998).

- Battery:

Battery maintenance depends on the level of use and the ambient conditions. The battery manufacturer's instructions must be obeyed.

The relevant manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

This Maintenance Schedule may include components which are not installed on your engine; these may be disregarded.

Out-of-service periods

If the engine is to remain out of service for more than 1 month, carry out engine preservation procedures in accordance with the Fluids and Lubricants Specifications, MTU Publication No. A001061.

Application group

4B Continuous power, medium duty

Load profile

Load-profile table

Load factor	100%	75%	50%	25%
Corresponding operating time	10%	40%	30%	20%

2 Maintenance schedule matrix

0-10.500 Operating hours

Item	Limit, years	Operating hours [h]																							
		Daily	500	1.000	1.500	2.000	2.500	3.000	3.500	4.000	4.500	5.000	5.500	6.000	6.500	7.000	7.500	8.000	8.500	9.000	9.500	10.000	10.500		
Lubrication points	1																								
Emergency air shut-off flaps	1																								
Engine oil filter	2																								
Engine operation	-	X																							
Fuel filter	2		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Valve gear	-			X		X		X		X		X		X		X		X		X		X			
Crankcase breathers	-							X						X						X					
Air filters	3							X						X						X					
Fuel injectors	-													X											
Combustion chambers	2													X											
Turbochargers	-																			X					
Component maintenance	-																			X					
Fuel delivery pump	-																			X					
Cylinder heads	-																			X					
Extended component maintenance	18																								

11.000-18.000 Operating hours

Item	Limit, years	Operating hours [h]																	
		11.000	11.500	12.000	12.500	13.000	13.500	14.000	14.500	15.000	15.500	16.000	16.500	17.000	17.500	18.000			
Lubrication points	1																		
Emergency air shut-off flaps	1																		
Engine oil filter	2																		
Engine operation	-																		
Fuel filter	2	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X			
Valve gear	-	X		X		X		X		X		X		X		X			
Crankcase breathers	-			X					X							X			
Air filters	3			X					X							X			
Fuel injectors	-			X												X			
Combustion chambers	2			X												X			
Turbochargers	-															X			
Component maintenance	-															X			
Fuel delivery pump	-															X			
Cylinder heads	-															X			
Extended component maintenance	18															X			

3 Maintenance tasks

Qual- ifica- tion level	Interval		Item	Maintenance tasks	Task
	[h]	[a]			
QL1	-	1	Lubrication points	Apply lubricant at lubrication points.	W1053
QL1	-	1	Emergency air shut-off flaps	Check operation of emergency air shut-off flaps (electrical). .	W1024
QL1	-	2	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years). .	W1008
QL1	Daily	-	Engine operation	Check engine oil level .	W0500
				Carry out visual inspection of engine for general condition and leaks. .	W0501
				Inspect intercooler drain system (if fitted). .	W0502
				Inspect service indicator of air filter. .	W0503
				Check the inspection bores of the HP fuel pump. .	W0504
				Check relief bores of water pump(s). .	W0505
				Check for abnormal running noises, exhaust gas colour and vibration. .	W0506
				Drain off water and contamination from fuel prefilter (if fitted). .	W0507
				Check service indicator of fuel prefilter (if fitted). .	W0508
QL1	500	2	Fuel filter	Fit new fuel filter or new fuel filter insert. .	W1001
QL1	1.000	-	Valve gear	Check valve clearance. .	W1002
QL1	3.000	-	Crankcase breathers	Crankcase breathers: Fit new filters or filter inserts.	W1046
QL1	3.000	3	Air filters	Fit new air filters. .	W1005
QL1	6.000	-	Fuel injectors	Fit new fuel injectors. .	W1006
QL1	6.000	2	Combustion chambers	Inspect cylinder chambers using endoscope. .	W1011
QL3	9.000	-	Turbochargers	Overhaul turbochargers.	W1038
QL3	9.000	-	Component maintenance	Before starting maintenance work, drain the coolant and flush the cooling systems.	W2000
				Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and camshaft running surfaces. .	W2001
				Clean air ducting. .	W2002
				Clean intercoolers and inspect for leakage. .	W2003
				Fit new high-pressure sensor in fuel system. .	W2004
				Check engine coolant thermostat and fit new thermal actuator. .	W2006
				Check charge-air coolant thermostat and fit new thermal actuator. .	W2007
				Overhaul engine coolant preheating system (if fitted). .	W2008
				Overhaul starter. .	W2010
				Inspect vibration damper. .	W2011
				Overhaul charge air coolant pump. .	W2070

Qual- ifica- tion level	Interval		Item	Maintenance tasks	Task
	[h]	[a]			
				Fit new sealing materials for all components disassembled. . Overhaul engine coolant pump	W2062 W2110
QL3	9.000	-	Fuel delivery pump	Fit new fuel delivery pump. .	W1051
QL3	9.000	-	Cylinder heads	Overhaul cylinder heads	W1134
QL4	18.000	18	Extended component maintenance	Completely disassemble the engine. Inspect engine components as per Maintenance Manual instructions and repair or fit new components as required. . Exchange all elastomeric parts and seals for new ones. . Fit new piston rings. . Fit new conrod bearings. . Fit new crankshaft bearings. . Fit new cylinder liners. . Fit new high-pressure fuel pump. . Fit new actuating cylinders for exhaust flow control flaps. . Fit new exhaust flow control flap bearings. . Fit new pressure relief valve in high-pressure fuel system. .	W3000 W3001 W3002 W3003 W3004 W3005 W3007 W3012 W3013 W3021